MODULE

# 01

# Networking Fundamentals

*Comprehensive Networking Course*

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## What You Will Learn

1. 1.1 What is Networking?
2. 1.2 Network Components
3. 1.3 Network Topologies
4. 1.4 Network Models (OSI & TCP/IP)

Mensah_Suku

SECTION 1.1

# What is Networking?

**Definition**

A computer network is a collection of interconnected devices that share resources and exchange data using common communication protocols.

## Share Resources

Printers, files, internet connections and storage shared efficiently across devices.

## Communicate

Email, video calls, chat and real-time collaboration across any distance.

## Centralize Data

Store data on servers — accessible, backed up and secured from one location.

GitHub.com/Riyazuddin

SECTION 1.1

# Real-World Use Cases

## Enterprise Office

Thousands of employees share files, printers, email and internet through a corporate LAN/WAN.

## E-Commerce

Online stores use networks to connect web servers, databases, payment gateways and CDN nodes.

## Healthcare

Patient records, medical imaging and remote diagnostics rely on secure, high-availability networks.

## Education

Distance learning platforms, shared libraries and campus Wi-Fi connect students and teachers.

## Smart Home

IoT devices — thermostats, cameras, speakers — communicate via a home LAN and the internet.

## Transportation

Airlines, logistics and GPS navigation use global networks for real-time tracking and routing.

SECTION 1.1

# Types of Networks

PAN
10m

## PAN

Personal Area Network

**~10 meters**

Connects personal devices around an individual. Bluetooth headsets, smartwatches, wireless keyboards.

*Bluetooth phone ↔ laptop*

LAN
1 km

## LAN

Local Area Network

**Building / Campus**

Connects devices within a single building or campus. High speed (1–10 Gbps). Controlled by one organisation.

*Office with 50 PCs on one switch*

MAN
50 km

## MAN

Metropolitan Area Network

**City-wide**

Spans a city or large campus. Often used by ISPs or city governments to connect multiple LANs.

*University spread across a city*

WAN
Global

## WAN

Wide Area Network

**Country / Global**

Connects networks across large distances — countries or continents. The Internet is the world's largest WAN.

*Corporate HQ ↔ branch offices worldwide*

SECTION 1.2

# Network Components

*Every network is built from three categories of physical and logical building blocks*

## End Devices

The sources and destinations of data on the network. These are the devices users interact with directly

EXAMPLES

*PCs, laptops, servers, smartphones, IP phones, printers, IoT sensors*

## Intermediary Devices

Forward, filter and direct traffic between end devices. They form the infrastructure of the network.

EXAMPLES

*Routers, switches, hubs, firewalls, wireless access points, modems*

## Transmission Media

The physical or wireless channels through which data travels between devices.

EXAMPLES

*Copper cable (Cat5e/Cat6), fiber optic, coaxial cable, Wi-Fi (radio waves)*

SECTION 1.2

# End Devices

*End devices initiate or receive communication — they are the network's users*



Workstation

## Desktop PC

Primary computing device for office workers. Connects via Ethernet or Wi-Fi.



Mobile Worker

## Laptop

Portable computing with Wi-Fi and sometimes cellular connectivity.



Service Host

## Server

Always-on machine providing files, email, web, DNS, DHCP services to clients.



Mobile Device

## Smartphone

Connects via cellular (LTE/5G) and Wi-Fi. Acts as both client and mini-server.



Shared Device

## Network Printer

Connected directly to the network. Receives print jobs from multiple users.



IoT / Security

## IP Camera

Streams video over the network. Each has its own IP address. Managed centrally.

Gin Sukut

SECTION 1.2

# Intermediary Devices

*These devices form the backbone — directing, filtering and forwarding traffic*

## Router

**Layer 3 — Network**

Connects different networks together. Makes intelligent path decisions using IP addresses and routing tables. The gateway to the internet.

## Switch

**Layer 2 — Data Link**

Connects devices within the same network. Uses MAC addresses to forward frames to the correct port. Modern switches support VLANs and PoE.

## Firewall

**Layer 3–7 — Multi**

Controls incoming and outgoing traffic based on security rules. Blocks unauthorised access and filters malicious content. First line of defence.

## Access Point

**Layer 2 — Wireless**

Provides wireless connectivity to the wired network. Manages SSIDs, channels and client associations. Enterprise APs are centrally managed.

## Hub (Legacy)

**Layer 1 — Physical**

Broadcasts data to all connected devices. Replaced by switches in modern networks due to inefficiency and collision domains. Rarely used today.

## Modem

**Layer 1–2**

Modulates/demodulates signals between your ISP and router. Converts digital packets to the signal type your ISP carries (cable, DSL, fibre).

SECTION 1.2

# Transmission Media

*The physical channels that carry data between devices*

## Twisted Pair Copper

Cat5e / Cat6 / Cat6a

**Speed:** Up to 10 Gbps

**Range:** Up to 100 metres

**Advantages**

Cheap, easy to install, widely available

**Limitations**

Limited range, susceptible to interference (EMI)

*Use case: Office LAN, home networking*

## Fiber Optic

Single-mode / Multi-mode

**Speed:** Up to 100+ Gbps

**Range:** Up to 80 km (SMF)

**Advantages**

Very high speed, immune to EMI, long range

**Limitations**

Expensive, fragile, requires specialist installation

*Use case: Data centres, ISP backbones, WAN links*

## Wireless (Wi-Fi)

802.11 a/b/g/n/ac/ax

**Speed:** Up to 9.6 Gbps (Wi-Fi 6)

**Range:** Up to ~150 m indoors

**Advantages**

No cables needed, flexible, mobility

**Limitations**

Shared medium, interference, security risks

*Use case: Mobile devices, IoT, public areas*

SECTION 1.3

# Network Topologies

*A topology describes how devices are physically or logically arranged and interconnected*

Network topology affects performance, fault tolerance, cost and scalability. Choosing the right topology is one of the first network design decisions.

**Star** | **Bus** | **Ring** | **Mesh** | **Hybrid**

| Topology | Central Device | Fault Tolerance | Cost | Common Use |
|---|---|---|---|---|
| **Star** | Required (switch) | High | Moderate | Office LANs, home networks |
| **Bus** | None (shared) | Low | Low | Legacy Ethernet (rarely used) |
| **Ring** | None | Medium | Moderate | SONET, Token Ring (legacy) |
| **Mesh** | None | Very High | High | WAN, military, data centres |
| **Hybrid** | Varies | High | High | Enterprise networks, internet |

Glinting

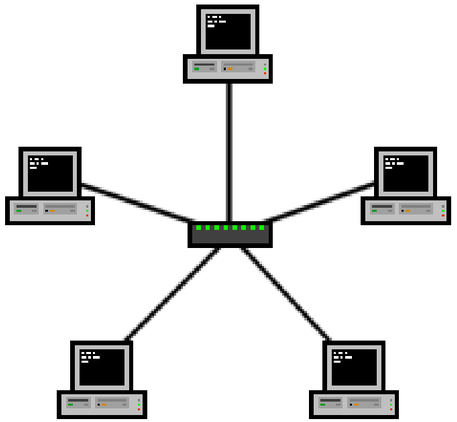
SECTION 1.3

# Star & Bus Topologies

*The two most historically important topologies in networking*

## STAR

*Most common modern topology*

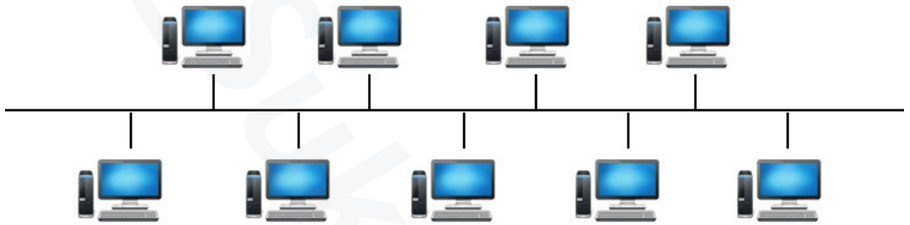


**Pros** Easy to add devices. One failure doesn't affect others. Easy troubleshooting.

**Cons** Central switch = single point of failure. More cable needed than bus.

## BUS

*Legacy — rarely used today*



**Pros** Simple, low cost. Easy initial setup. No central device required.

**Cons** One cable break disrupts ALL devices. Slow when busy — shared medium.

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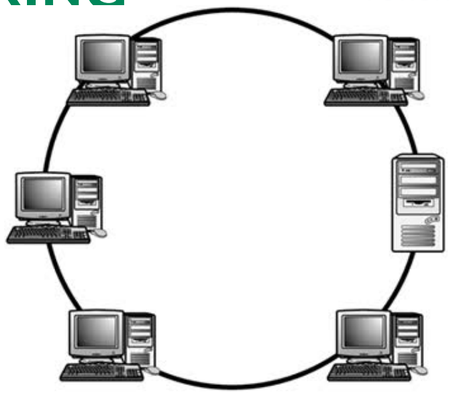
SECTION 1.3

# Ring & Mesh Topologies

*Understanding fault tolerance trade-offs through topology design*

## RING

*Token Ring / SONET*



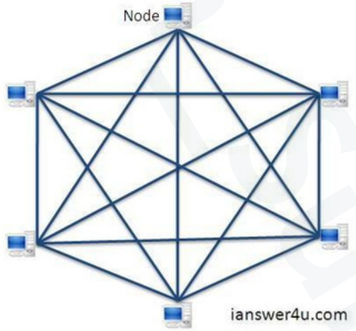
Each device connects to exactly two others, forming a closed loop. Data travels in one direction (or both in dual-ring). A break in the ring can disrupt the whole network unless dual-ring is used.

**Pros** Predictable performance. Equal access. No collision domain.

**Cons** One break stops the network (single ring). Adding devices disrupts the ring.

## MESH

*Full Mesh / Partial Mesh*




Every device connects to every other device (full mesh) or to multiple devices (partial mesh). Extremely fault-tolerant — multiple paths exist between any two nodes. Used in WANs and data centres.

**Pros** Maximum redundancy. No single point of failure. Multiple paths.

**Cons** Very expensive — n(n-1)/2 connections needed. Complex to manage.

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# Hybrid Topology

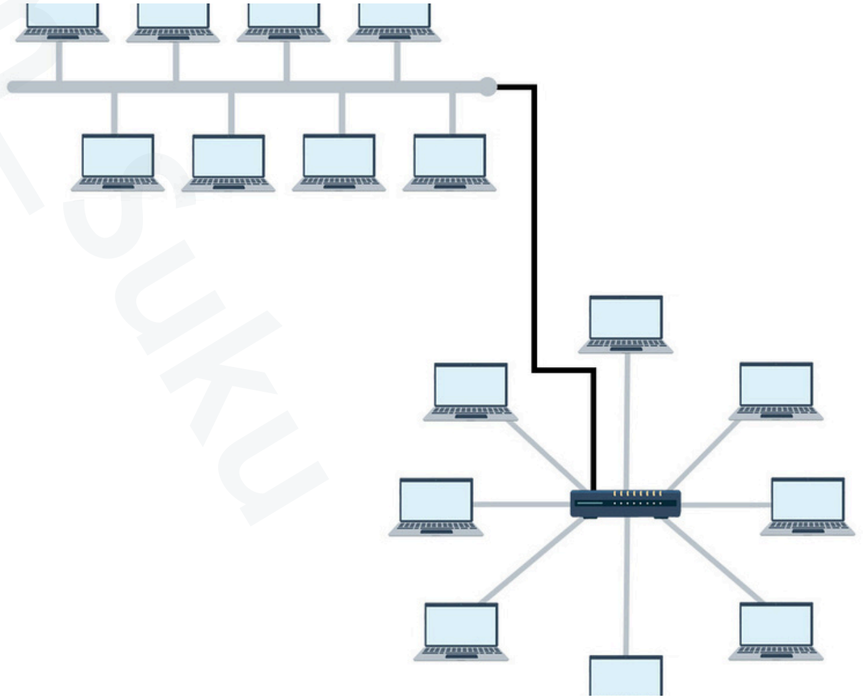
*Most real-world networks combine multiple topologies to balance cost, performance and reliability*

A hybrid topology uses two or more topology types in the same network. Enterprise networks nearly always use hybrids — star for access layer, mesh for core layer

**Core Layer**
Mesh — high redundancy between core routers

**Distribution**
Partial mesh between distribution switches

**Access Layer**
Star — end devices connect to access switches



### Scalable
Grow each layer independently without redesigning the whole network.

### Resilient
Redundant paths at the core mean no single failure disrupts the entire network.

### Cost-Managed
Use expensive full-mesh only where needed; star for cheap access-layer wiring.

SECTION 1.4

# Network Models

*Frameworks that define how network communication is structured and standardised*

Network models provide a common language for describing how data moves from one device to another. Two models dominate: the OSI Model (theoretical) and the TCP/IP Model (practical).

## OSI Model

Open Systems Interconnection

**Layers:** 7 Layers

**Origin:** ISO Standard (1984)

**Purpose:** Theoretical framework for teaching

Conceptual — defines what each layer does but does not dictate how it should be implemented. Universal reference model used worldwide for understanding and troubleshooting.

## TCP/IP Model

Transmission Control Protocol / IP

**Layers:** 4 Layers

**Origin:** DARPA / DoD (1970s)

**Purpose:** Practical model used on the internet

Implementation-based — maps directly to the protocols actually used on the internet. Simpler and more practical than OSI. The model that the internet was built on.

SECTION 1.4

# The OSI Model — 7 Layers

*Each layer has a specific role. Data passes through all 7 layers when sent and received*

| | | | | |
|---|---|---|---|---|
| 7 | **Application** | Data | Interface between user applications and the network. | *HTTP, FTP, SMTP, DNS, DHCP* |
| 6 | **Presentatio** | Data | Data translation, encryption/decryption, compression. | *SSL/TLS, JPEG, ASCII, MPEG* |
| 5 | **Session** | Data | Establishes, manages and terminates communication sessions. | *NetBIOS, RPC, SQL sessions* |
| 4 | **Transport** | Segment | Reliable delivery, flow control, error recovery, ports. | *TCP, UDP* |
| 3 | **Network** | Packet | Logical addressing (IP), routing between networks. | *IP, ICMP, OSPF, RIP* |
| 2 | **Data Link** | Frame | MAC addressing, frame delivery on local network. | *Ethernet, Wi-Fi (802.11), PPP* |
| 1 | **Physical** | Bits | Transmits raw bits over physical medium. | *Ethernet cable, fiber, Wi-Fi radio* |

## Memory Aid

Top → Bottom:

*"All People*
*Seem To*
*Need Data*
*Processing"*

Bottom → Top:

*"Please*
*Do Not*
*Throw*
*Sausage*
*Pizza Away"*

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SECTION 1.4

# TCP/IP Model & OSI Comparison

*How the practical TCP/IP model maps to the theoretical OSI reference model*

## Application

OSI 5, 6, 7

Handles formatting, sessions and application data.

*HTTP, FTP, SMTP, DNS, DHCP, SSH*

## Transport

OSI 4

Reliable or fast delivery. Ports, segmentation, flow control.

*TCP, UDP*

## Internet

OSI 3

Logical addressing and routing across networks using IP.

*IP, ICMP, ARP, OSPF*

## Network Access

OSI 1, 2

Physical transmission and local frame delivery using MAC.

*Ethernet, Wi-Fi, PPP*

## Key Differences

| OSI Model | TCP/IP |
|---|---|
| 7 layers | 4 layers |
| Theoretical / academic | Practical / deployed |
| Protocol-independent | Built around TCP & IP |
| Session & Presentation | Merged into Application |
| Used for teaching | Used for implementation |

In practice: Engineers use OSI to troubleshoot and TCP/IP to build. Both are essential knowledge for the CCNA exam.

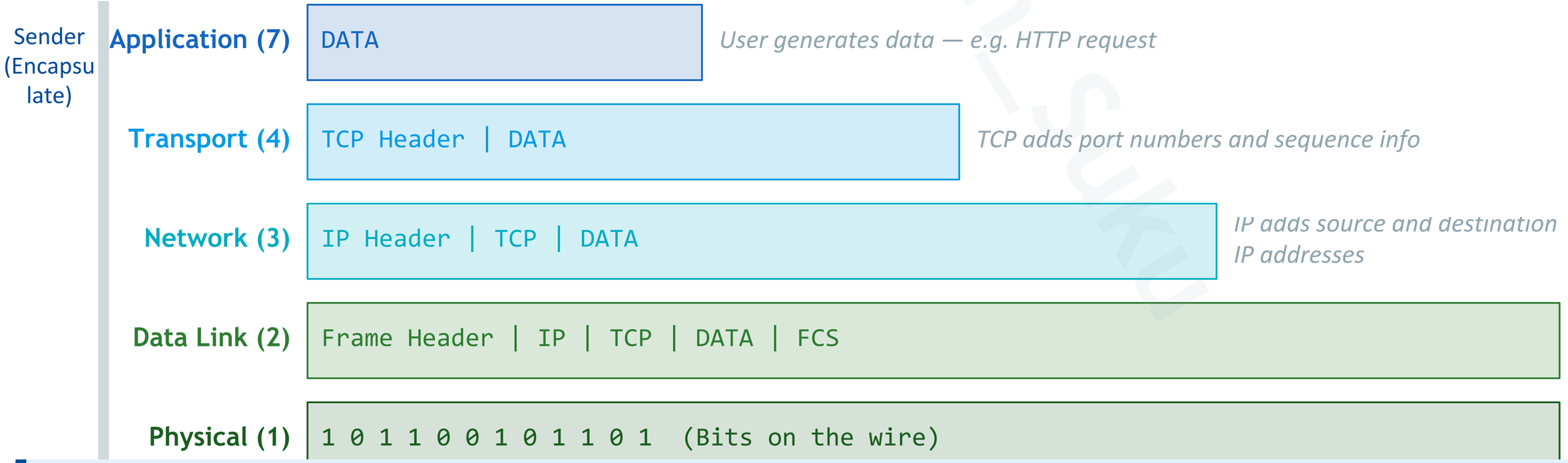
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SECTION 1.4

# Data Encapsulation & De-encapsulation

*How data is wrapped with headers (and trailers) as it moves down the OSI stack*

Encapsulation adds layer-specific headers as data moves DOWN the OSI model. De-encapsulation removes them as data moves UP on the receiving device.




Key Rule: Each layer only communicates with the same layer on the other device (peer-to-peer logic), even though data physically passes through all layers.

Gini Sukul

# Networking Fundamentals — Key Takeaways

1.1

## What is Networking?

- Connects devices to share resources and communicate
- LAN, WAN, MAN, PAN — classified by geographic scale
- Powers enterprise, healthcare, education and smart homes

1.2

## Network Components

- End devices: source & destination of data
- Intermediary: routers, switches, firewalls, APs
- Media: copper, fibre, wireless — each with trade-offs

1.3

## Network Topologies

- Star: most common — easy to troubleshoot
- Bus: legacy, shared medium, prone to failure
- Mesh: maximum redundancy, high cost
- Hybrid: used in all real enterprise networks

1.4

## Network Models

- OSI: 7 layers — theoretical, used for troubleshooting
- TCP/IP: 4 layers — practical, used on the internet
- Encapsulation adds headers as data moves down the stack

KNOWLEDGE CHECK

# Module 1 Review Questions

*Test your understanding of networking fundamentals before moving to Module 2*

**Q1**

**What is the main difference between a LAN and a WAN?**

*Hint: Think about geographic scope and who controls the infrastructure*

**Q2**

**Name the 7 layers of the OSI model in order from bottom to top.**

*Hint: Physical → Data Link → Network → Transport → Session → Presentation → Application*

**Q3**

**A company needs to connect offices in 3 different countries. What network type is this?**

*Hint: Consider geographic scope — building? city? country?*

**Q4**

**Why is fiber optic cable preferred over copper for long-distance connections?**

*Hint: Think about speed, range and interference*

**Q5**

**What is the difference between a switch and a router?**

*Hint: Which OSI layer does each operate at? What addresses do they use?*

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